



EMMANUEL
COLLEGE

ALUMNI DATA PROJECT

Anisa Koni

Lena Karaguesian

-Under the guidance of Professor. Aishwariya Elangovan

Questions



Which states are the most alumni located in?



Where would it be best to hold future alumni events?

Project Objective

- Analyze alumni distribution across the U.S. to identify optimal locations for future alumni events using R-based data visualization.

Dataset Overview



Dataset includes alumni location, graduation year, and state distribution.



Total records: ~16,000.



Focus: Geographic clustering of alumni.

Tools & Techniques



- R Programming



- Data Cleaning



- Choropleth Maps



- Distribution Analysis



- Exploratory Data Analysis (EDA)

Alumni per State Data

Alumni_Data

State	Alumni_count
Alabama	11
Alaska	5
Arizona	87
Arkansas	5
California	456
Colorado	88
Connecticut	1017
Delaware	24
Florida	585
Georgia	85
Hawaii	21
Idaho	9
Illinois	91

Indiana	23
Iowa	15
Kansas	4
Kentucky	11
Louisiana	27
Maine	396
Maryland	173
Massachusetts	2000
Michigan	42
Minnesota	34
Mississippi	5
Missouri	24
Montana	14

South Carolina	89
South Dakota	4
Tennessee	35
Texas	155
Utah	15
Vermont	146
Virginia	169
Washington	75
West Virginia	11
Wisconsin	17
Wyoming	3

Nebraska	4
Nevada	22
New Hampshire	802
New Jersey	226
New Mexico	37
New York	566
North Carolina	149
North Dakota	0
Ohio	63
Oklahoma	16
Oregon	39
Pennsylvania	125
Rhode Island	605

Choropleth Code using R

```
Alumni_Data <- read.csv("Alumni_Data_NoMA.csv")
usa <- st_as_sf(maps::map("state", fill=TRUE, plot=FALSE)) %>%
  sf::st_transform(crs=4236)
usa
Alumni_Data %>% mutate(ID=tolower(State)) -> Alumni_Data
alum_map_data <- left_join(x=usa, y=Alumni_Data, by="ID")
ggplot(alum_map_data) + geom_sf(aes(fill=Alumni_count)) +
  scale_fill_viridis(option="cividis") +
  labs(title=paste("Alumni per State")) + theme_bw()
```

Steps

1. Load alumni data

```
Alumni_Data <- read.csv("Alumni_Data_NoMA.csv")
```

2. Gather US map data

```
usa <- st_as_sf(maps::map("state", fill=TRUE, plot=FALSE)) %>%  
  sf::st_transform(crs=4236)
```

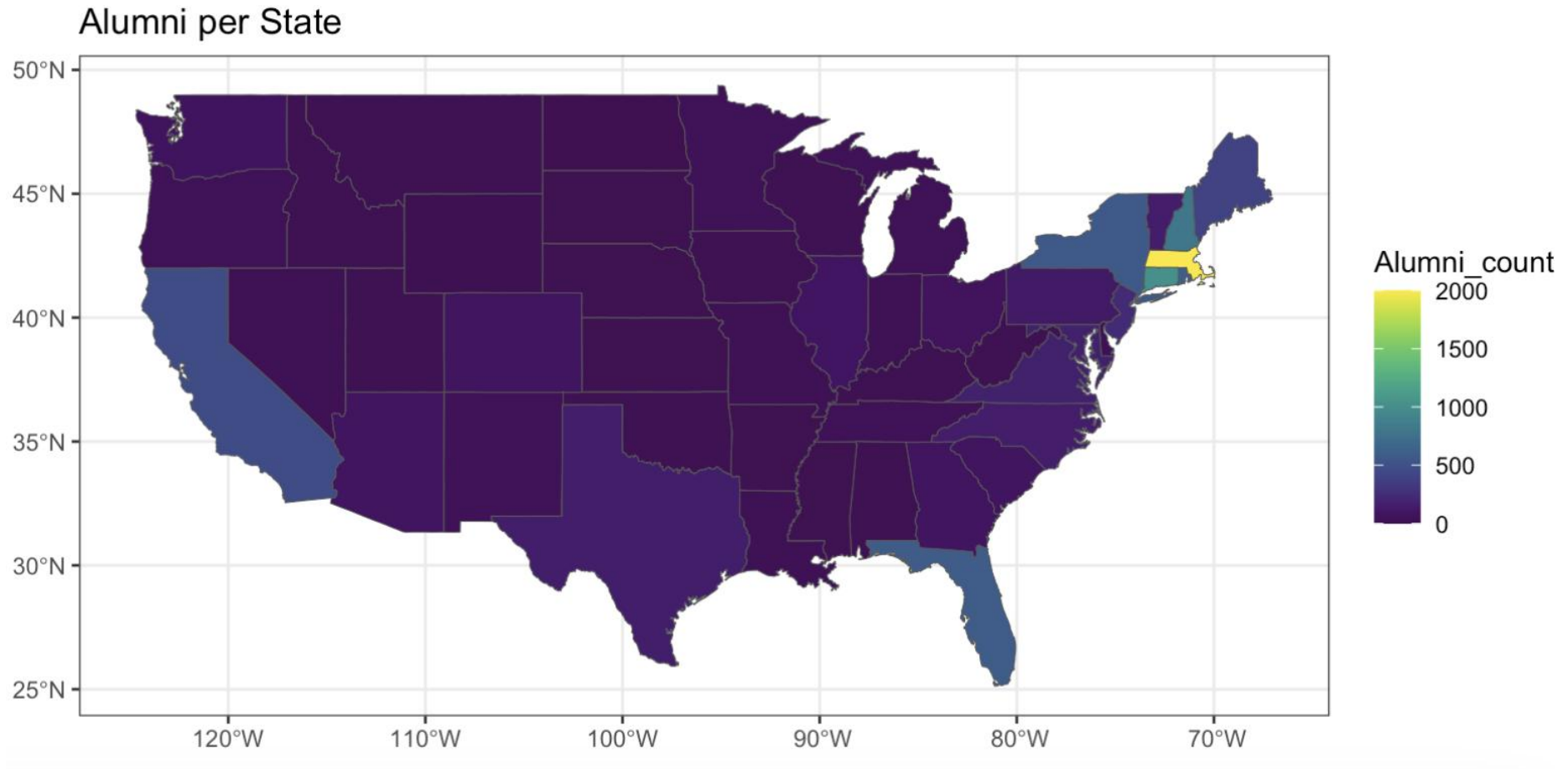
3. Join alumni and US map data

```
Alumni_Data %>% mutate(ID=tolower(State)) -> Alumni_Data  
alum_map_data <- left_join(x=usa, y=Alumni_Data, by="ID")
```

4. Plot alumni data on US map

```
ggplot(alum_map_data) + geom_sf(aes(fill=Alumni_count)) +  
  scale_fill_viridis(option="cividis") +  
  labs(title=paste("Alumni per State")) + theme_bw()
```

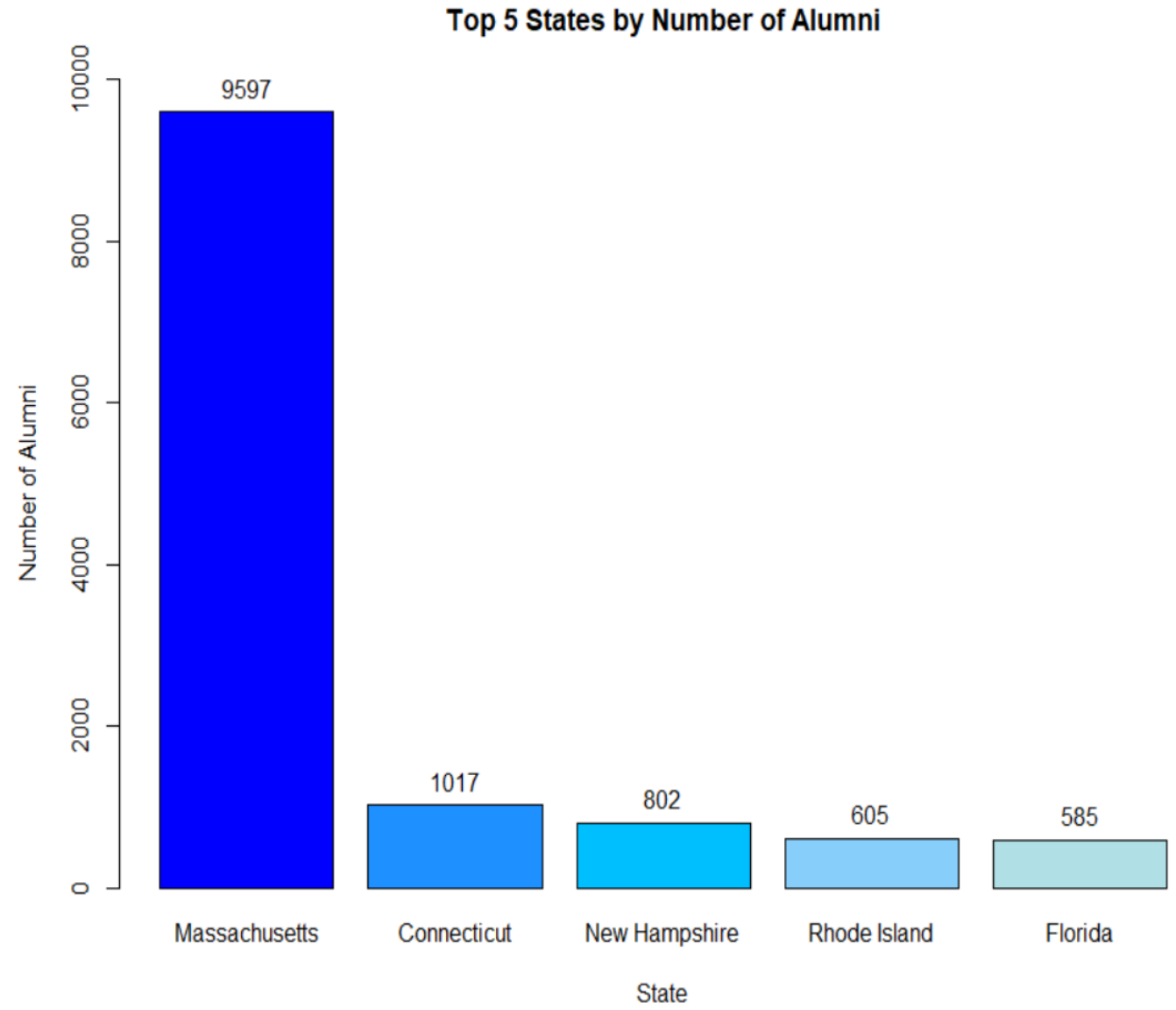

Choropleth Map



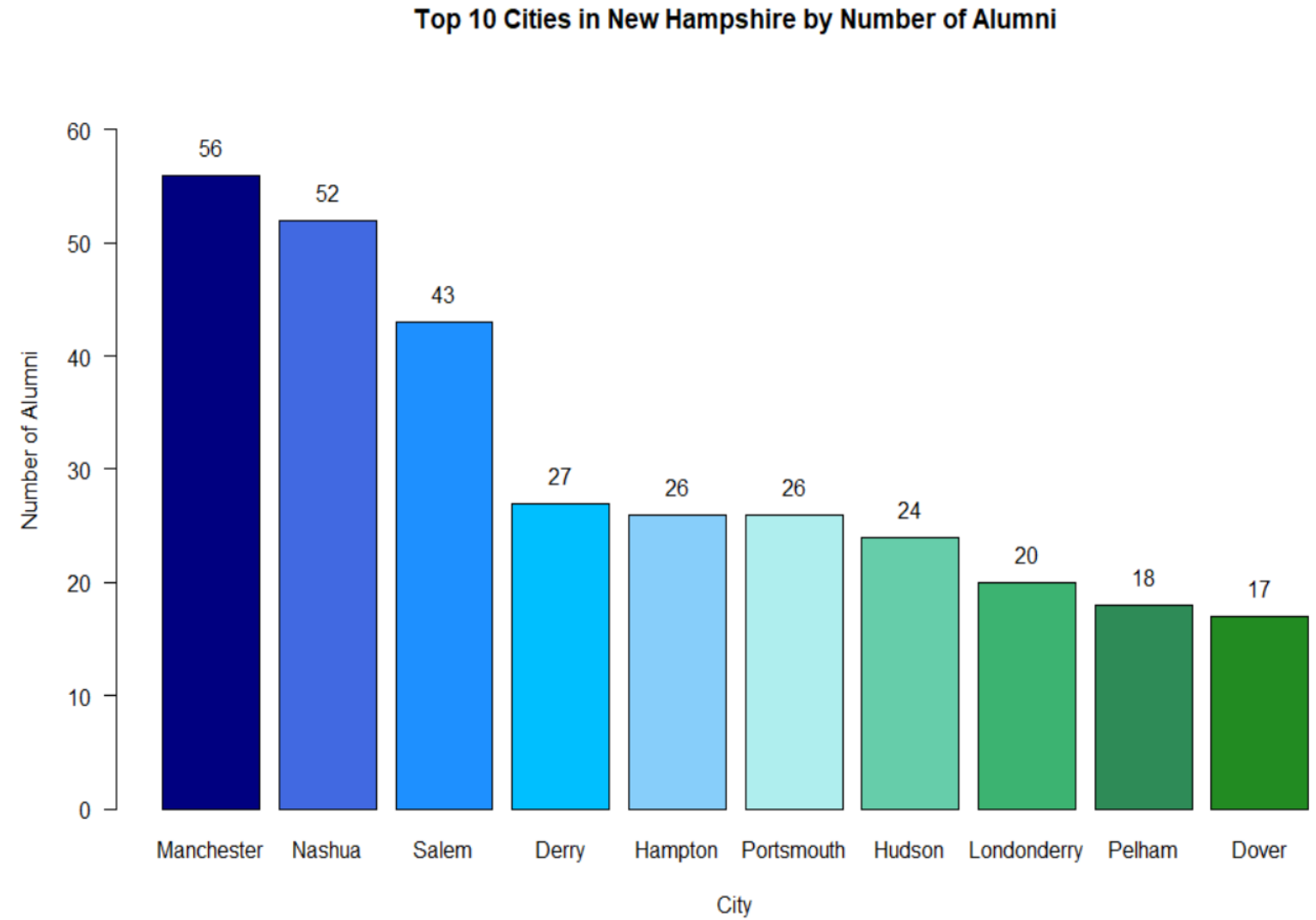
Map Results

- Massachusetts has the highest number of alumni (about 10,000 alumni)
- The following most populated states are Connecticut and New Hampshire (about 1000 alumni)
- Other highly populated states are New York, Florida, and California (about 500 alumni)
- Based on these results, we did an exploratory analysis of alumni in New Hampshire

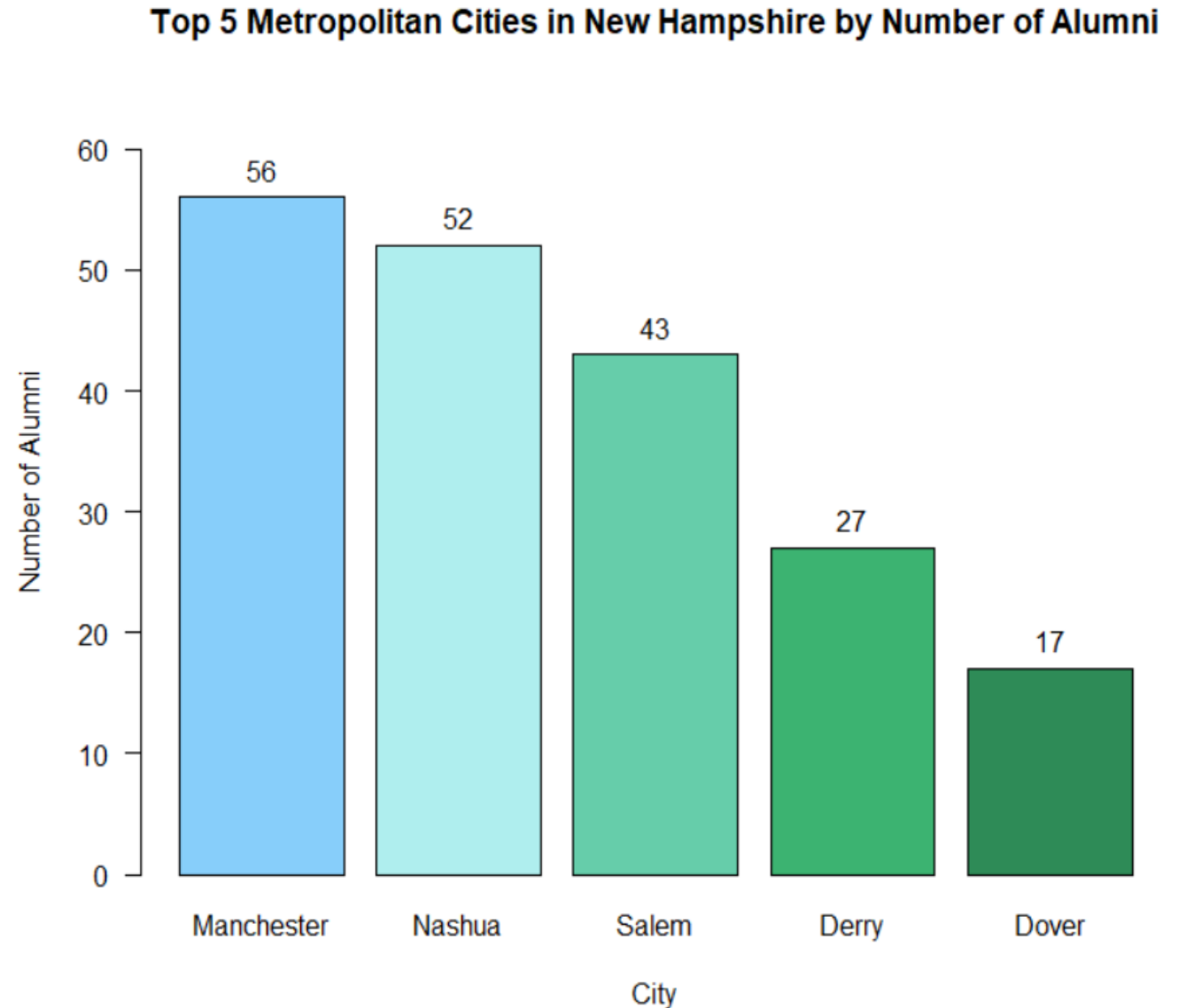
- Top 5 states where alumni are located
- Decided to look more into NH since there is a high population of alumni



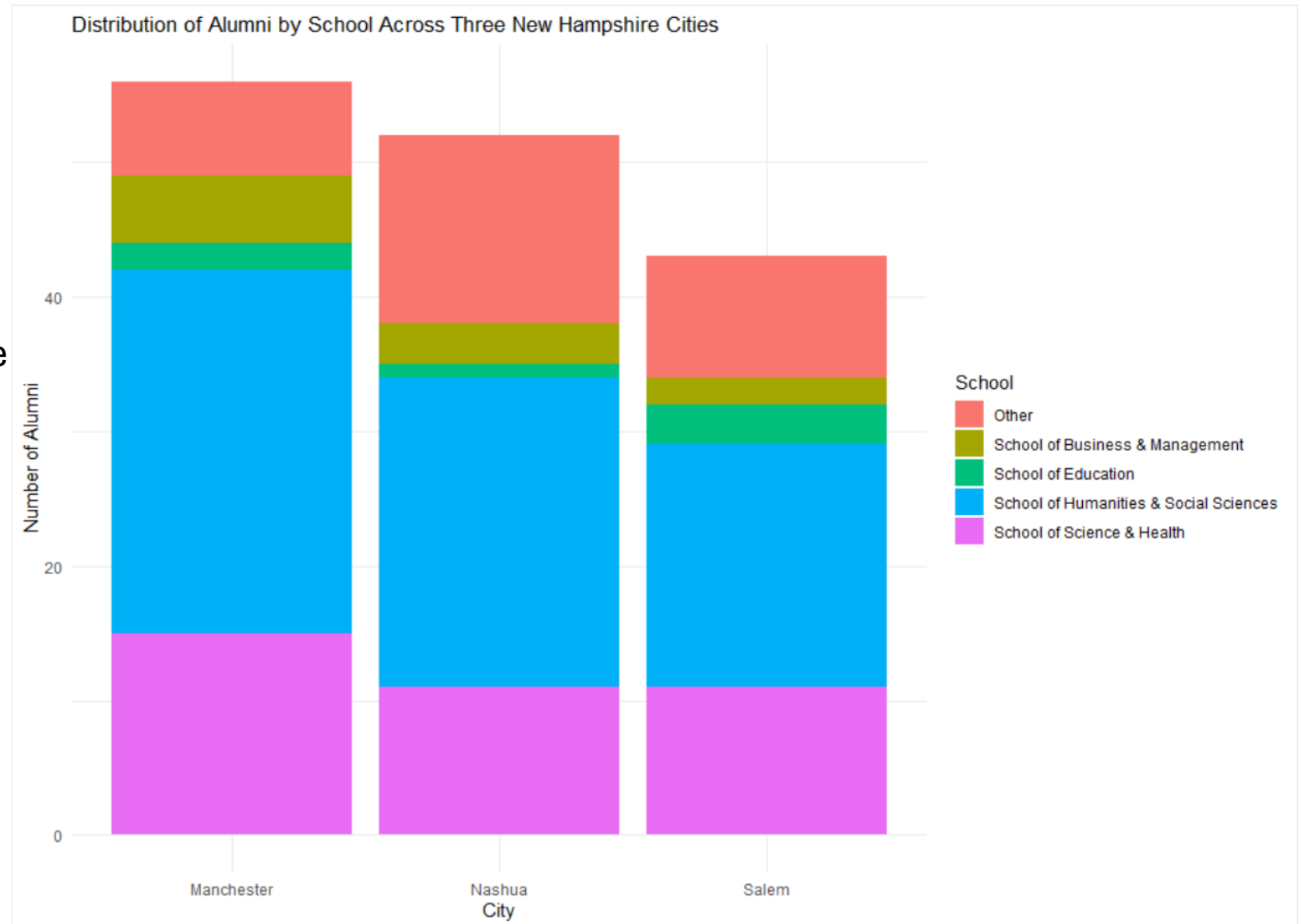
- Looked at the distribution of alumni across cities in NH to see possible locations for an alumni event



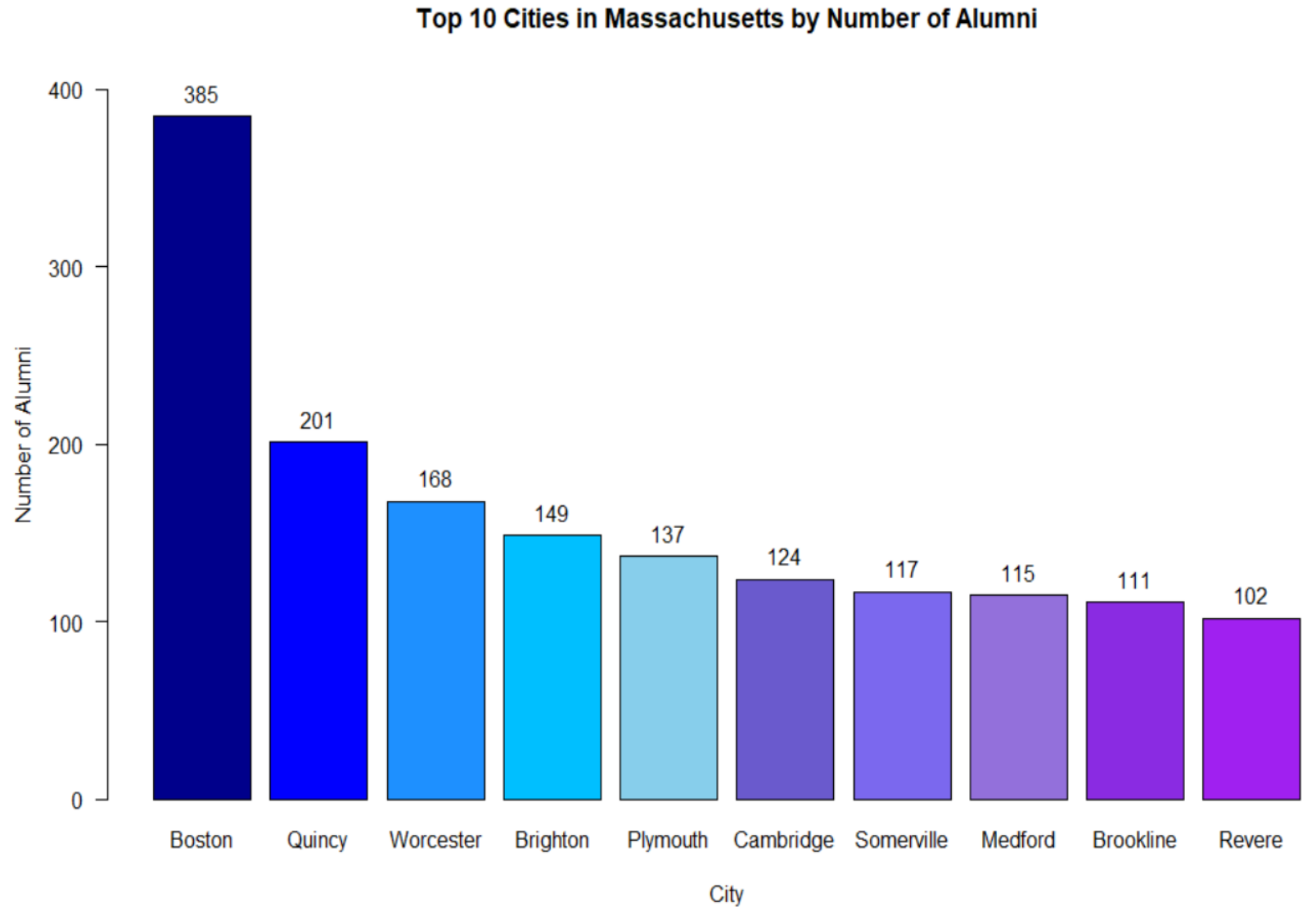
- Looked at the largest cities in NH to see which city would be best to do an alumni event
- The top 4 largest cities (Manchester, Nashua, Salem, Derry) are the same for the distribution of alumni



- Looked at the number of alumni in each school
- No school of nursing since there are no nursing majors in NH



- Looked at the top 10 cities in MA to see possible locations for an alumni event



Results

There are about 800 alumni in NH

A little under 50% of alumni in NH live in Manchester, Nashua, and Salem

A little over 50% of alumni in NH in Manchester, Nashua, and Salem have a major within the School of Humanities and Social Sciences

Conclusion

- New Hampshire could be a possible location for an alumni event and in one of the top 3 cities which are Manchester, Nashua, and Salem
- Since there are many alumni in NH with majors in the School of Humanities and Social Sciences, a targeted alumni event could be held here